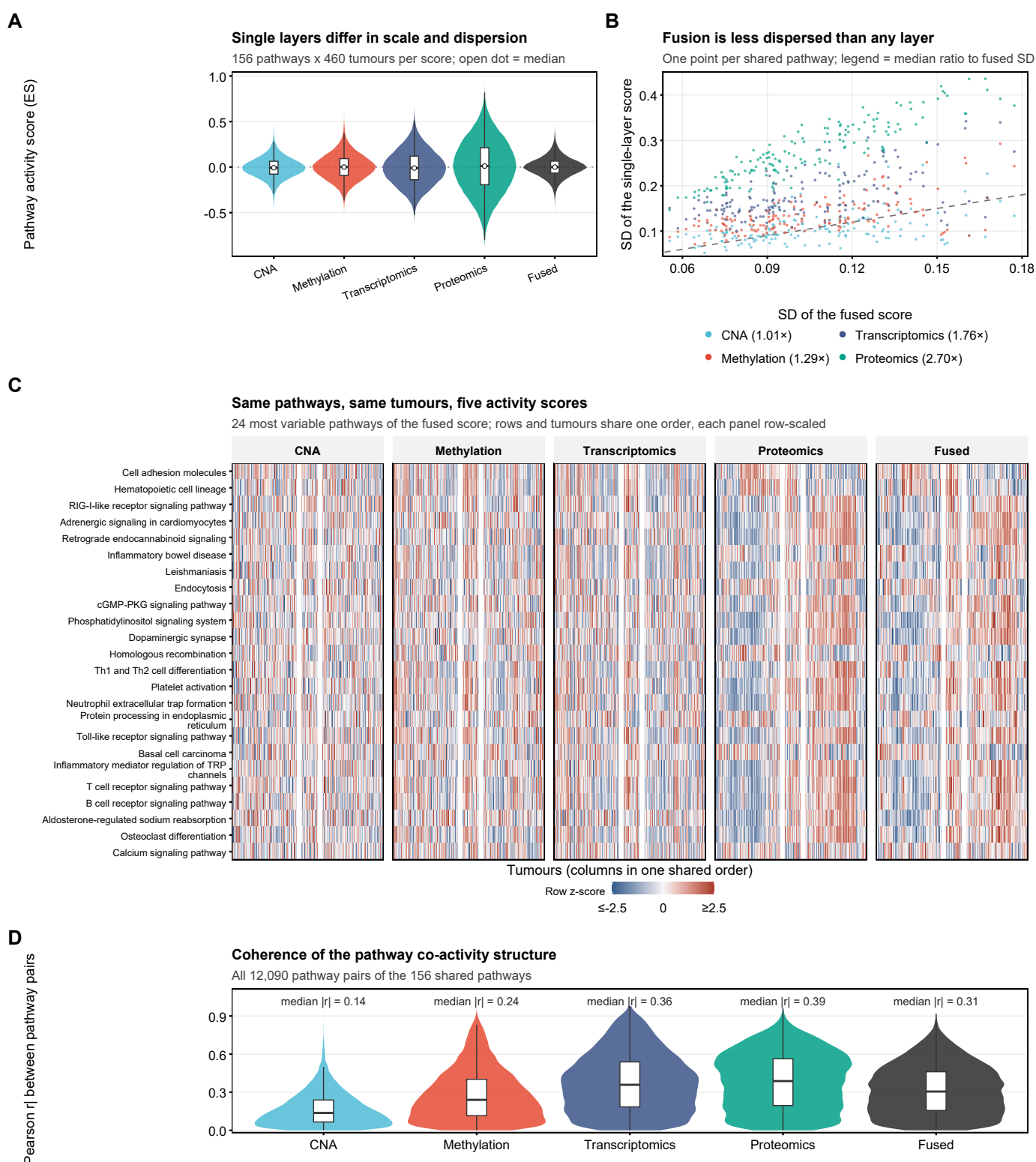

A
Single layers differ in scale and dispersion
156 pathways x 460 tumours per score; open dot = median
Pathway activity score (ES)
1.0
0.5
0.0
-0.5
CNA
Methylation
Transcriptomics
Proteomics
Fused
B
Fusion is less dispersed than any layer
One point per shared pathway; legend = median ratio to fused SD
SD of the single-layer score
0.4
0.3
0.2
0.1
0.06
0.09
0.12
0.15
0.18
SD of the fused score
CNA (1.01×)
Transcriptomics (1.76×)
Methylation (1.29×)
Proteomics (2.70×)
C
Same pathways, same tumours, five activity scores
24 most variable pathways of the fused score; rows and tumours share one order, each panel row-scaled
CNA
Methylation
Transcriptomics
Proteomics
Fused
Cell adhesion molecules
Hematopoietic cell lineage
RIG-I-like receptor signaling pathway
Adrenergic signaling in cardiomyocytes
Retrograde endocannabinoid signaling
Inflammatory bowel disease
Leishmaniasis
Endocytosis
cGMP-PKG signaling pathway
Phosphatidylinositol signaling system
Dopaminergic synapse
Homologous recombination
Th1 and Th2 cell differentiation
Platelet activation
Neutrophil extracellular trap formation
Protein processing in endoplasmic reticulum
Toll-like receptor signaling pathway
Basal cell carcinoma
Inflammatory mediator regulation of TRP channels
T cell receptor signaling pathway
B cell receptor signaling pathway
Aldosterone-regulated sodium reabsorption
Osteoclast differentiation
Calcium signaling pathway
Tumours (columns in one shared order)
Row z-score
≤-2.5
0
≥2.5
D
Coherence of the pathway co-activity structure
All 12,090 pathway pairs of the 156 shared pathways
Pearson r| between pathway pairs
median |r| = 0.14
median |r| = 0.24
median |r| = 0.36
median |r| = 0.39
median |r| = 0.31
0.9
0.6
0.3
0.0
CNA
Methylation
Transcriptomics
Proteomics
Fused